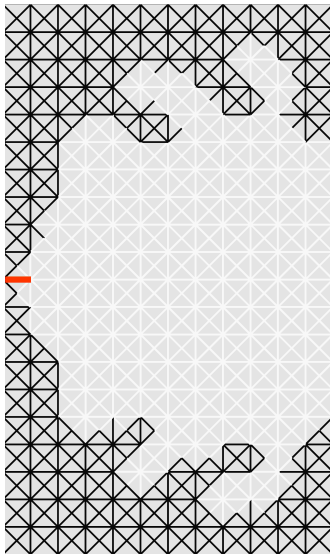


$N = 1408$

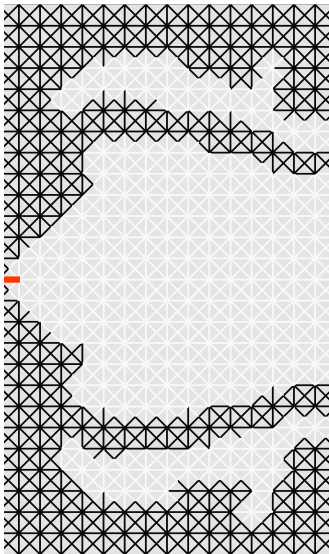


$$Q/Q_{\text{lb}}^{\text{TM}} = 1.061$$

$$Q_{\text{E}}/Q_{\text{lb}}^{\text{TM}} = 4 \times 10^{-4}$$

a)

$N = 2454$



$$Q/Q_{\text{lb}}^{\text{TM}} = 1.057$$

$$Q_{\text{E}}/Q_{\text{lb}}^{\text{TM}} = 4 \times 10^{-4}$$

b)